CORONARY HEART DISEASE (CHD) AND DIABETES MELLITUS

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INTENDED LEARNING OUTCOMES

- **By the end of the lecture, students should be able to:**
  - **Know the Magnitude of the CAD and DM problem**
  - **Identify both modifiable and non-modifiable risk factors of CAD and DM.**
  - **Know the preventive measures of CAD and DM.**

CORONARY HEART DISEASE (CHD)

- **Definition**: It is the **impairment of the function** of the heart due to **inadequate blood flow** to the myocardium, as a result of obstruction in the coronary circulation (atherosclerosis, plaque, thrombus).
- Coronary artery disease (CAD) is **manifested** in any of the following forms:

  *Angina pectoris of effort * Myocardial infarction

  *Irregularities of the heart *Cardiac failure

  *Cardiac arrest

MAGNITUDE OF THE PROBLEM:

- (CAD) is the leading cause of death
  - 25 to 30 percent of total deaths in developed countries.
  - 15% of deaths in the developing countries
- CAD is considered as 'modern epidemic
  - 2 to 3 times greater in urban areas than rural areas.
  - one and a half times more among men than among women.
  - maximum in the age group of 50 to 60 years.

RISK FACTORS:

Non-modifiable Risk Factors

- Age (>50 years, peak at 50–60 years)
- Male sex
- Family history
- Genetic predisposition (cholesterol, LDL regulation)

MODIFIABLE RISK FACTORS

- Smoking (major risk, dose-dependent)
- Hypertension (accelerates atherosclerosis)
- High serum cholesterol (>220 mg/dl)
- Diabetes mellitus (2–3 times risk)
- Obesity (linked to LDL, hypertension, diabetes)
- Physical inactivity (low HDL, increase obesity/ hypertension)
- Hormones (oral contraceptives, hyperestrogenemia)
- Type A personality (stress-prone)
- Alcohol (heavy use)
- Environmental factors: soft water, chronic noise (>110 dB)
- Drugs (fenfluramine, phentermine misuse)
- High serum homocysteine (>15.5 µmol/L)
- Other biochemical factors: hyperuricemia, increase ferritin, increase ceruloplasmin

PREVENTION OF CORONARY ARTERY DISEASE

Primary Prevention:

Purpose

- Limit the incidence of disease by controlling causes and risk factors.
- Depends on population-wide changes to reduce average risk.

1. Dietary Changes

- Saturated fats: <10% of total energy intake.
- Cholesterol intake: <300 mg/day/adult.
- Serum cholesterol: <200 mg/dl.
- Increase carbohydrate consumption proportionately.

2. Smoking Control

- Goal: Smoke-free society.
- Requires education, legislation, and public programs.

3. Blood Pressure Management

- Even small reductions lower CAD incidence.
- Strategies: low salt diet, reduced alcohol, physical activity, weight control.

4. Physical Activity

- Regular exercise prevents obesity, hypertension, and indirectly CAD.

5. Genetic Predisposition

- Early detection of conditions (e.g., familial hypercholesterolemia, hypertriglyceridemia, childhood hypertension, obesity) for timely prevention.

6. Mental Health Promotion

- Encourage relaxed lifestyle, recreation, stress reduction.

7. Control of Oral Contraception

8. Management of Pathological Risk Factors

- Prevention & control of diabetes, hypertension, obesity.

9. Health Education

- Promote group sports, exercise, healthy work conditions, family relations, personal satisfaction.

- **Secondary Prevention**
  - stops the progress of the disease in its early stage and prevents complications.
  - This consists of prevention of recurrence of CAD by cessation of exposure to risk factors, e.g.
    - giving up smoking after an attack of myocardial infarction, followed by bypass surgery,
    - prompt treatment with antihypertensives after detecting hypertension,
    - similarly starting antidiabetic drugs after detection of diabetes mellitus, etc.

DIABETES MELLITUS

- **Greek: Diabetes → "Siphon" (passing water).**
- **Latin: Mellitus → "Honey" (sweet).**
- **Definition:**
  - **A metabolic syndrome characterized by: Polyuria , Polyphagia , Polydipsia , Hyperglycemia , Glycosuria**
  - **Caused by absolute or relative deficiency of insulin (by secretion, action, or both).**
- **Insulin regulates metabolism of carbohydrates, proteins, fats, and electrolytes.**
- **Insulin deficiency:**
  - **Absolute deficiency → Insulin-Dependent Diabetes Mellitus (IDDM).**
  - **Relative/partial deficiency → Non-Insulin-Dependent Diabetes Mellitus (NIDDM).**

CLASSIFICATION OF DIABETES MELLITUS

1. PRIMARY

- **Type 1 (insulin dependent diabetes mellitus) (IDDM).**
  - Absolute insulin deficiency
  - Juvenile onset (<30 years)
  - Sudden onset, thin build, autoimmune association, hla-linked
  - Acute complications, family history usually absent
- **Type 2 (Noninsulin dependent diabetes mellitus) (NIDDM)**
  - Relative insulin deficiency
  - Maturity onset (>30 years)
  - Gradual onset, obesity-related, lifestyle-linked
  - Chronic complications, family history common

SECONDARY

- Pancreatic pathology (e.g., pancreatitis, tumors, cystic fibrosis)
- Other causes: hemochromatosis, trauma, autoimmunity
- Hormonal excess (catecholamines, GH, glucagon, glucocorticoids)
- Long-term drug use (corticosteroids, thiazides, OCPs)
- Liver disease, genetic syndromes

MAGNITUDE OF THE PROBLEM

- Diabetes mellitus is an 'Iceberg disease'. It is an ubiquitous malady of the world today, affecting about 150 million people worldwide.
- As of 2025, about 589 million adults (aged 20–79 years) worldwide are living with diabetes.
- 4th leading cause of death in the USA.
- Rising in younger populations due to urbanization, industrialization, lifestyle changes.
- Impaired Glucose Tolerance (IGT): Intermediate state, prevalence 3.6–9.1%.

- The diagnostic criteria (OGTT):
  - Fasting blood sugar >125 mg/dL
  - 2-hour post-glucose load >180 mg/dL

COMPLICATIONS OF DIABETES

- **Acute**
  - **Hypoglycemia**
  - **Hyperglycemia & ketoacidosis**
  - **Infections (fungal, urinary, anaerobic, mycobacterial)**
- **Chronic**
  - **Macrovascular: Atherosclerosis → cerebrovascular & coronary artery disease**
  - **Microvascular: Retinopathy, nephropathy, neuropathy**
  - **Foot ulcers & infections**
  - **Increased risk of ischemic heart disease (3.5–4 times higher)**
  - **Often coexists with hypertension, dyslipidemia, obesity, metabolic syndrome**

RISK FACTORS
NON-MODIFIABLE RISK FACTOR

1. Age

- Diabetes can occur in any age group.
- Type 1 is common among younger age group below 30 and type 2 among middle aged and elderly.
- Thus, age constitutes a risk-factor for DM specially type 2
- Younger the age, worse is the prognosis.

2. Sex

- Type 1 DM is common among men and type 2 among women.

3. Genetic Factors

- Type 2 shows 90 percent concordance genetic component whereas type 1 shows only 50 percent.
- Maturity onset diabetes of the young (MODY) is an autosomal dominant hereditary disease. If both the parents are such diabetics, the children have 100 percent chance of developing it.
- Type 1 has HLA (human leukocyte antigen) linked genetic predisposition, whereas type 2 is not associated with HLA.

MODIFIABLE RISK FACTOR

1. Diet

- **High-risk dietary patterns:**
  - Excess sugar, saturated fats, and traditional "conservative diets" (butter, potatoes, whole milk) increase risk of type 2 DM.
- **Protective dietary patterns:**
  - "Prudent diet" rich in fruits and vegetables.
  - Diets high in non-starch polysaccharides (NSP), such as cereals and whole grains.

2. Sedentary Lifestyle

- Lack of physical activity combined with overnutrition contributes to obesity.
- Strongly linked to type 2 DM due to reduced insulin sensitivity.

3. **Obesity –**

- Central adiposity (waist circumference, waist-to-hip ratio) is a stronger predictor than BMI.
- Increases insulin resistance and/or reduces insulin receptor numbers on target cells.
- Obesity plays no role in Type 1 DM.

4. **Stress and strain:**

- Such as pregnancy, surgery, trauma and such other stressful situations may 'bring out' the disease.

5. **Alcoholism:**

- Excessive intake of alcohol can increase the risk of type 2 DM by damaging the liver and pancreas by promoting obesity.

OTHER FACTORS:

- **1. Socioeconomic class:**
  - Prevalence of DM is no longer related to socioeconomic status.
- **2. Immunological factors:**
  - Type 1 DM is a slow autoimmune disease, occurring over many years.
- **3. Viral infections:**
  - The important viruses that can result in the development of DM are Rubella, Mumps, Human coxsackie B4, Encephalomyocarditis (EMC) virus, Mengo virus 2T and Rheo virus type I. They tend to result in type 1 DM.
- **4. Pregnancy:**
  - Can cause gestational diabetes; repeated pregnancies increase risk of permanent DM.

PRIMARY PREVENTION

- Through elimination or modification of risk factors. The main points are:
  - Improvements in the nutritional habits (such as avoiding sweets and fatty foods, alcohol, diabetogenic drugs like oral pills, steroids, regular and adequate intake of proteins, high intake of dietary fibers)
  - Maintenance of body weight by doing moderate exercise.
  - Correction of obesity
  - Avoiding over nutrition and alcohol

- **Changing lifestyle**
- **Regular exercises**
- **Maintenance of normal body weight**
- **Avoidance of oral contraceptives and steroids**
- **Reduction of factors promoting atherosclerosis such as hypertension, smoking, cholesterol level, etc.**
- **Yoga exercises and meditation goes a long way in the prevention and control of diabetes mellitus.**

SECONDARY PREVENTION

- It is by early diagnosis and prompt treatment of DM.
  - Early diagnosis is done by screening procedures.
- The aims of treatment are:
  - To maintain normal blood glucose level
  - To maintain normal body weight.

- The principles of treatment are:
- Diet alone
  - small balanced meals more frequently
  - No fast and no feast.
  - More of raw vegetables, less of cereals.
- Diet and oral antidiabetic drugs,
  - i.e. restriction of the diet associated with oral hypoglycemic drugs for type 2 DM.
- Diet and insulin,
  - i.e. restriction of diet and administration of insulin injection daily for Type 1 DM.
- The ultimate objective of the management of DM is to prevent complications

TERTIARY PREVENTION

- **Disability limitation:**
  - This is to limit the development of further disability in a diabetic person, who comes late with complications.
  - This consists of giving an intensive treatment in diabetic clinics.
- **Rehabilitation:**
  - This is given for those who have become disabled and handicapped due to complications resulting in blindness, amputation of leg, etc.
  - They are rehabilitated physically, mentally, socially, psychologically and vocationally.

Thank you

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